PROFESSIONAL EDUCATION



SUMMER 2024

DEEP LEARNING FOR AI AND COMPUTER VISION

DATES	DURATION	FORMAT	FEE	CEUS
Summer 2024	4 Days	On Campus	$3,760	2.5

Explore the cutting edge of technological research in a field that is poised to transform the world

Acquire the skills you need to build advanced computer vision applications featuring innovative developments in neural network research. Designed for engineers, scientists, and professionals in healthcare, government, retail, media, security, and automotive manufacturing.

Deep learning innovations are driving exciting breakthroughs in the field of computer vision. Robots and drones not only "see", but respond and learn from their environment. Autonomous cars avoid collisions by extracting meaning from patterns in the visual signals surrounding the vehicle.

This course covers the latest developments in vision AI, with a sharp focus on advanced deep learning methods, specifically convolutional neural networks, that enable smart vision systems to recognize, reason, interpret and react to images with improved precision.

EARN A PROFESSIONAL CERTIFICATE

This course may be taken individually or as part of the ***Professional Certificate Program in Machine Learning & Artificial Intelligence.***

WHO SHOULD ATTEND

This course is designed for professionals from industries such as healthcare, government, security, and automotive manufacturing, who are looking to solve computer vision problems with deep learning. Participants who will benefit from the curriculum include:

- **Engineers** looking to design and build advanced computer vision applications
- **Data scientists** who want to explore cutting-edge developments in neural network research and deep learning models, and acquire strategies for applying them to their work
- **Managers and business executives** seeking to capitalize on the latest advancements in deep learning for computer vision and AI
- **Government and military officials** who need to understand how computer vision applications are deployed in pertinent technologies, such as drones and other defense systems

professional.mit.edu/cv

LEARNING OUTCOMES

Participants will explore the latest developments in neural network research and deep learning models that are enabling highly accurate and intelligent computer vision systems capable of understanding and learning from images. We will start from fundamental topics in image modeling, including image formation, feature extraction, and multiview geometry, then move on to the latest applications in object detection, 3D scene understanding, vision and language, image synthesis, and vision for embodied agents. By the end, participants will:

- Be familiar with fundamental concepts and applications in computer vision
- Grasp the principles of state-of-the art deep neural networks
- Understand low-level image processing methods such as filtering and edge detection
- Gain knowledge of high-level vision tasks such as object recognition, scene recognition, face detection and human motion categorization
- Develop practical skills necessary to build highly-accurate, advanced computer vision applications

LEAD INSTRUCTORS

Antonio Torralba



Antonio Torralba is a Professor of Electrical Engineering and Computer Science at the Massachusetts Institute of Technology (MIT), the MIT director of the MIT-IBM Watson AI Lab, and the inaugural director of the MIT Quest for Intelligence, a MIT campus-wide initiative to discover the foundations of intelligence.

Phillip Isola



Phillip Isola is an assistant professor in the MIT Department of Electrical Engineering and Computer Science. He studies visual intelligence from the perspective of both minds and machines. He received a National Science Foundation (NSF) graduate fellowship as well as an NSF postdoctoral fellowship.

Teaching Methodology

The certificate curriculum is grounded in the spirit of MIT's motto, "Mens et Manus," or "mind and hand," which combines theoretical instruction with hands-on methods of discovery.



INTERACTIVE LECTURES



HANDS-ON EXERCISES



GROUP PROJECTS



SUPPLEMENTAL RESOURCES

PROFESSIONAL EDUCATION

ABOUT MIT PROFESSIONAL EDUCATION

For 70 years, MIT Professional Education has been providing technical professionals worldwide a gateway to renowned MIT research, knowledge, and expertise, through advanced education programs designed specifically for them. In addition to industry-focused, two-to-five-day live virtual and on-campus courses through Short Programs, MIT Professional Education offers professionals the opportunity to take online and blended learning courses through Digital Plus Programs, attend courses abroad through International Programs, enroll in regular MIT academic courses through the Advanced Study Program, or attend Custom Programs designed specifically for their companies. For more information, please visit **professional.mit.edu**.